R13

Code No: 126AQ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, May - 2019
INFORMATION SECURITY
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a)	Specify the components of encryption algorithm.	[2]
b)	Mention the drawback of Steganography.	[3]
c)	Specify the applications of the public key cryptosystem.	[2]
d)	Specify the design criteria of block cipher.	[3]
e)	What is meant by Kerberos?	[2]
f)	What is the role of compression function in hash function?	[3]
g)	What are the benefits of IP Security?	[2]
h)	Explain the digital signature service provided by PGP.	[3]
i)	What are the limitations firewall?	[2]
j)	Discuss about virtual electrons.	[3]

PART - B

(50 Marks)

2.a) List and explain different types of Security Attacks.
b) Discuss about Network security model with neat illustration. [5+5]

OR

3.a) Explain the playfair cipher with example.
b) Explain the transposition technique. [5+5]

4.a) With a neat block diagram explain the single round of DES algorithm.
b) Discuss about various cipher block modes of operation. [5+5]

OR

5.a) Explain RSA algorithm.
b) In a RSA system, it is given that p = 11, q = 13, e = 7 and M = 5, Find ciphertext C and also M from decryption. [5+5]

6.a) Give the general format of X.509 certificate.
b) Give the design objectives for HMAC. [5+5]

OR

7.a) Give the requirements that must be met by authentication.
b) Discuss about Secure Hash Algorithm (SHA). [5+5]

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8.a) What do you mean by Security Association? Specify the parameters that identify the Security Association.

b) Explain the IP security architecture. [5+5]

OR

9. Explain PGP Message generation and PGP message reception techniques. [10]

10.a) Discuss about the attacks on Packet Filtering Firewall.

b) Discuss in detail about the Distributed Intrusion Detection. [5+5]

OR

11. Describe the following:
SET operation, SET features, Dual signature and payment processing. [10]

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